Idea Proposal

Istanbul Reservoir Foresight

Probabilistic Early Warning for Istanbul's Reservoirs

1. Project Idea

Istanbul Reservoir Foresight will forecast the combined storage of Istanbul's 10 drinking-water reservoirs up to 90 days ahead and estimate the probability that storage will fall below 30% of capacity within the next 60 days. The tool will help water planners prepare for shortages and compare how changes in consumption and water transfers affect that risk.

2. Relevance to the Sustainable Development Goals

Testing water-saving scenarios supports water-use efficiency (SDG 6.4). Modeling rainfall, consumption, and transfers together supports integrated water resources management (SDG 6.5). Earlier warnings give planners more time to prepare for drought (SDG 13.1).

3. Literature Examples

Akiner (2025, PeerJ) analyzed over 23 years of daily storage data from 10 Istanbul reservoirs to assess seasonal patterns and water management options. DOI 10.7717/peerj.20041

Badem et al. (2024, Sustainability) compared daily, weekly, 15-day, and 30-day forecasts for seven Istanbul reservoirs using historical storage, weather, consumption, and evapotranspiration data. DOI 10.3390/su16177696

4. Dataset

The Istanbul Metropolitan Municipality's open dataset covers about 8,500 days of reservoir-level records for 10 dams, from October 23, 2000, to February 19, 2024. We will use records from İSKİ and verifiable archives to compile more recent storage, consumption, and Melen and Yeşilçay transfer data. We will combine these records with ERA5 rainfall, temperature, and evapotranspiration data from Open-Meteo in a daily CSV, standardize units, remove duplicates, and flag missing or derived values.

5. Machine Learning Approach

We will use LightGBM to learn nonlinear relationships between daily storage changes and tabular inputs such as accumulated rainfall, consumption, and transfers. We will project storage 90 days ahead using a water balance model driven by ECMWF forecasts and historical rainfall scenarios. The resulting trajectories will provide a median, an 80% prediction interval, and a probability of falling below the threshold. In chronological tests, we will compare LightGBM with a persistence baseline and SARIMAX using mean absolute error at 30, 60, and 90 days, as well as alert accuracy and lead time.